5-stage-ROSC

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Intro

⇒ TSMC 65nm

⇒ Set Parameter

⇒ Calculate FoM

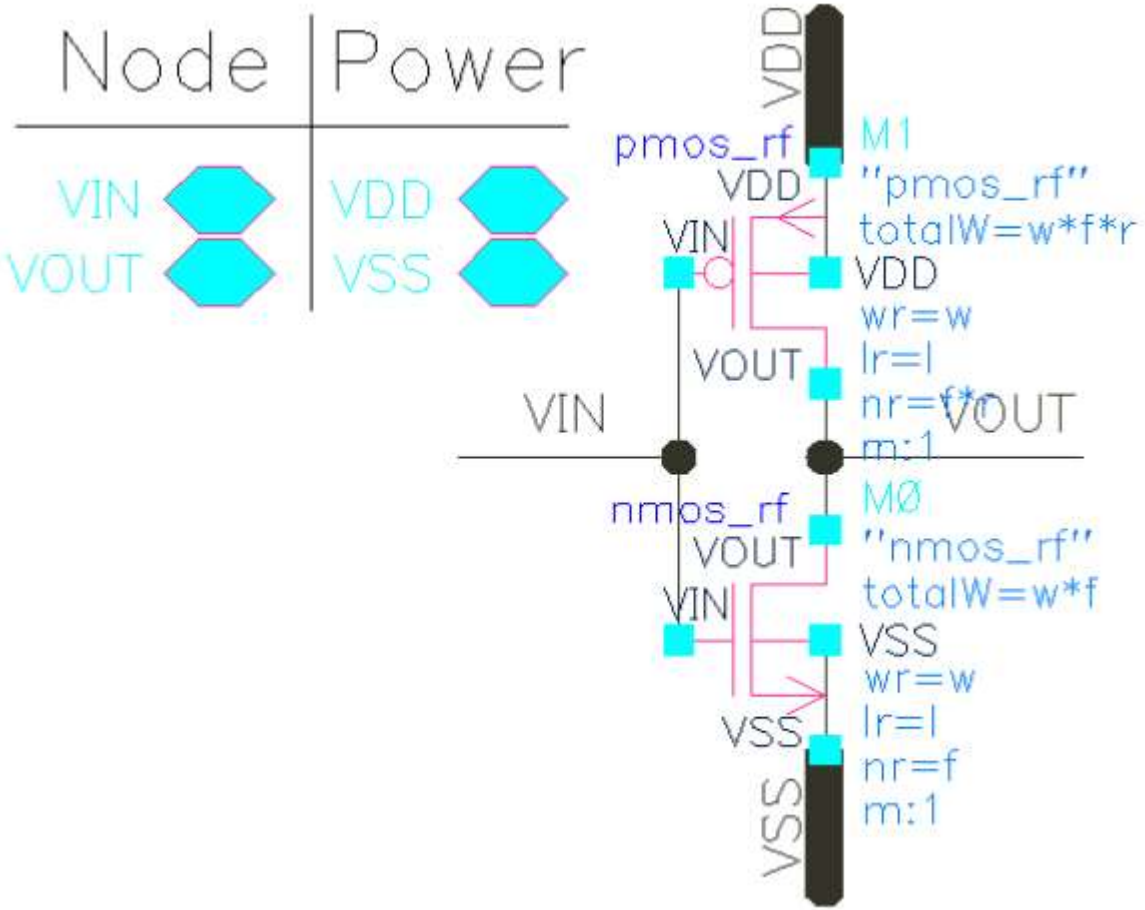
⇒ TSMC 180nm

⇒ Set Parameter

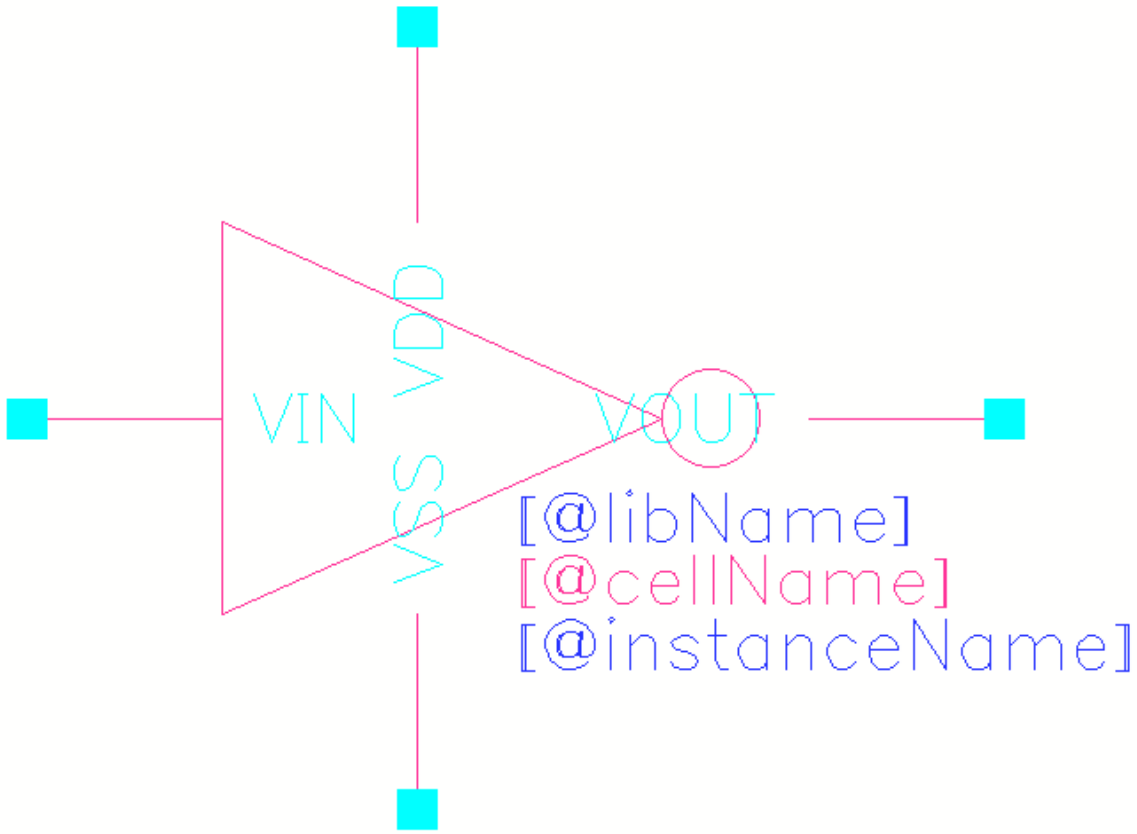
⇒ Calculate FoM

⇒ Comparison

65nm schematic

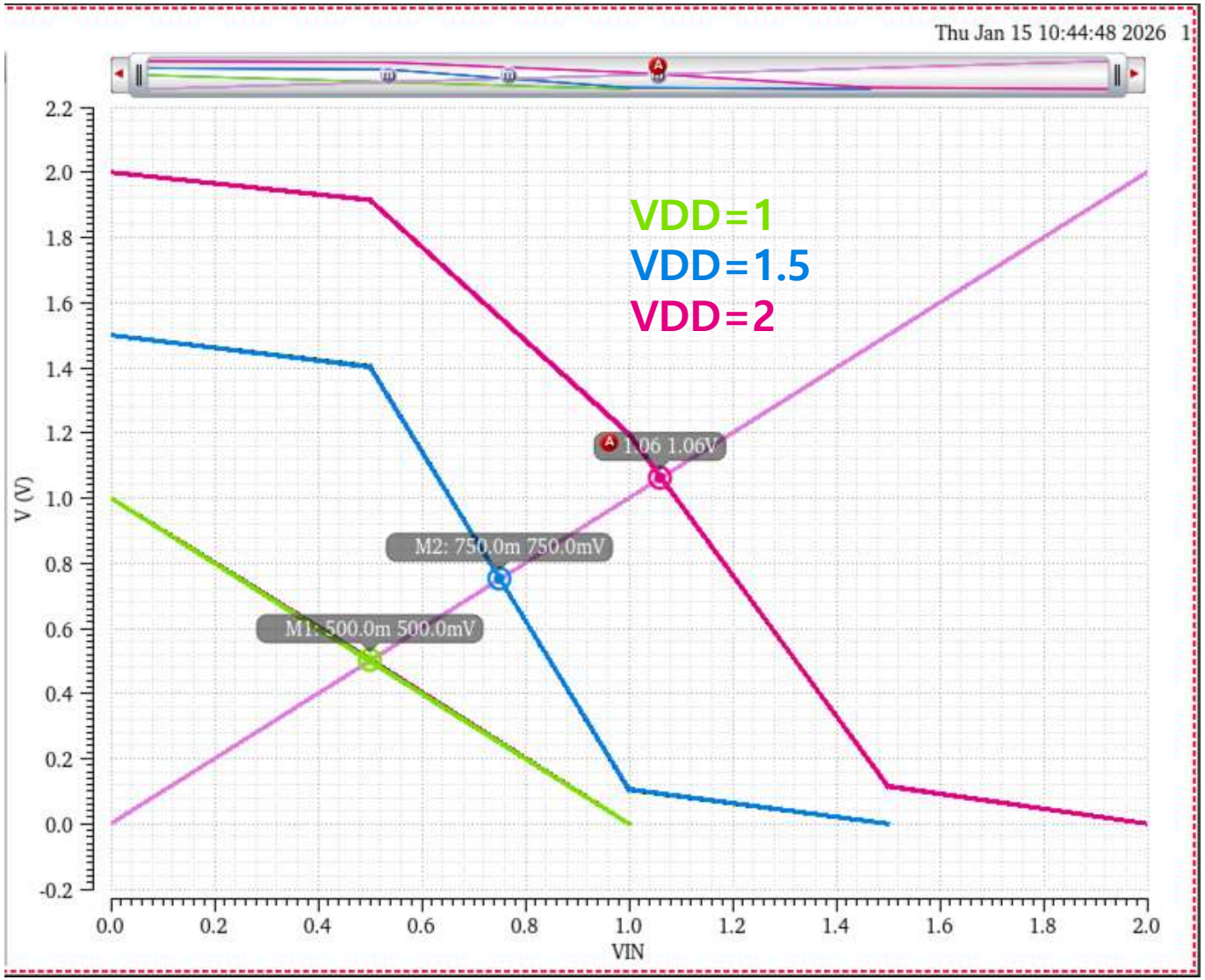


schematic



symbol

VDD



Logic Threshold Simulation

정상 동작을 위한 VDD=1V~1.5V

⇒ 구동 전압을 높이면 발진주파수가 올라감

⇒ 전류량 증가

⇒ 충방전 속도 향상

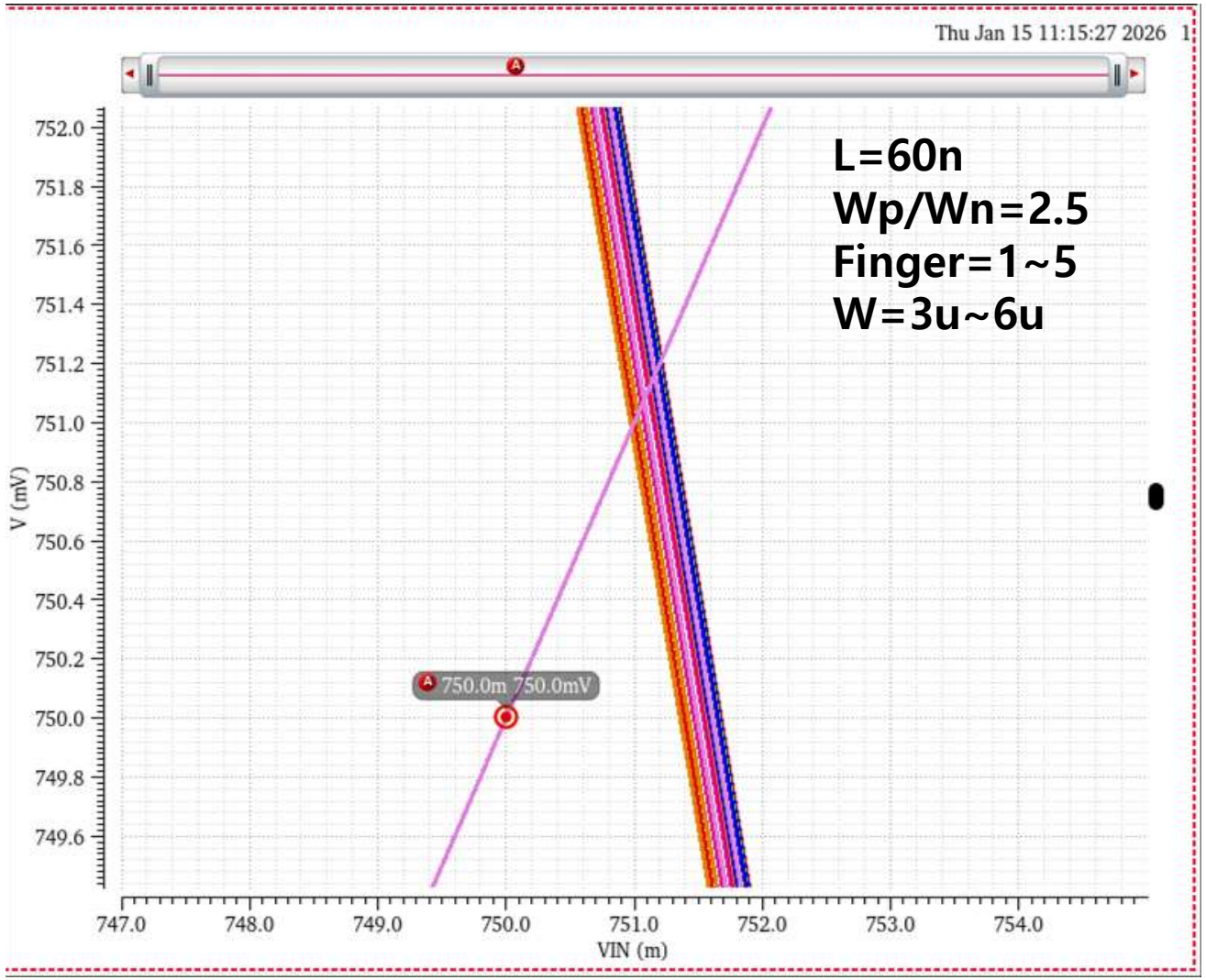
⇒ 지연 시간 감소

⇒ 안정적인 mosfet 동작을 위해

기준전압에서 마진 +20%까지

⇒ 구동 전압 1.2로 설정

Finger

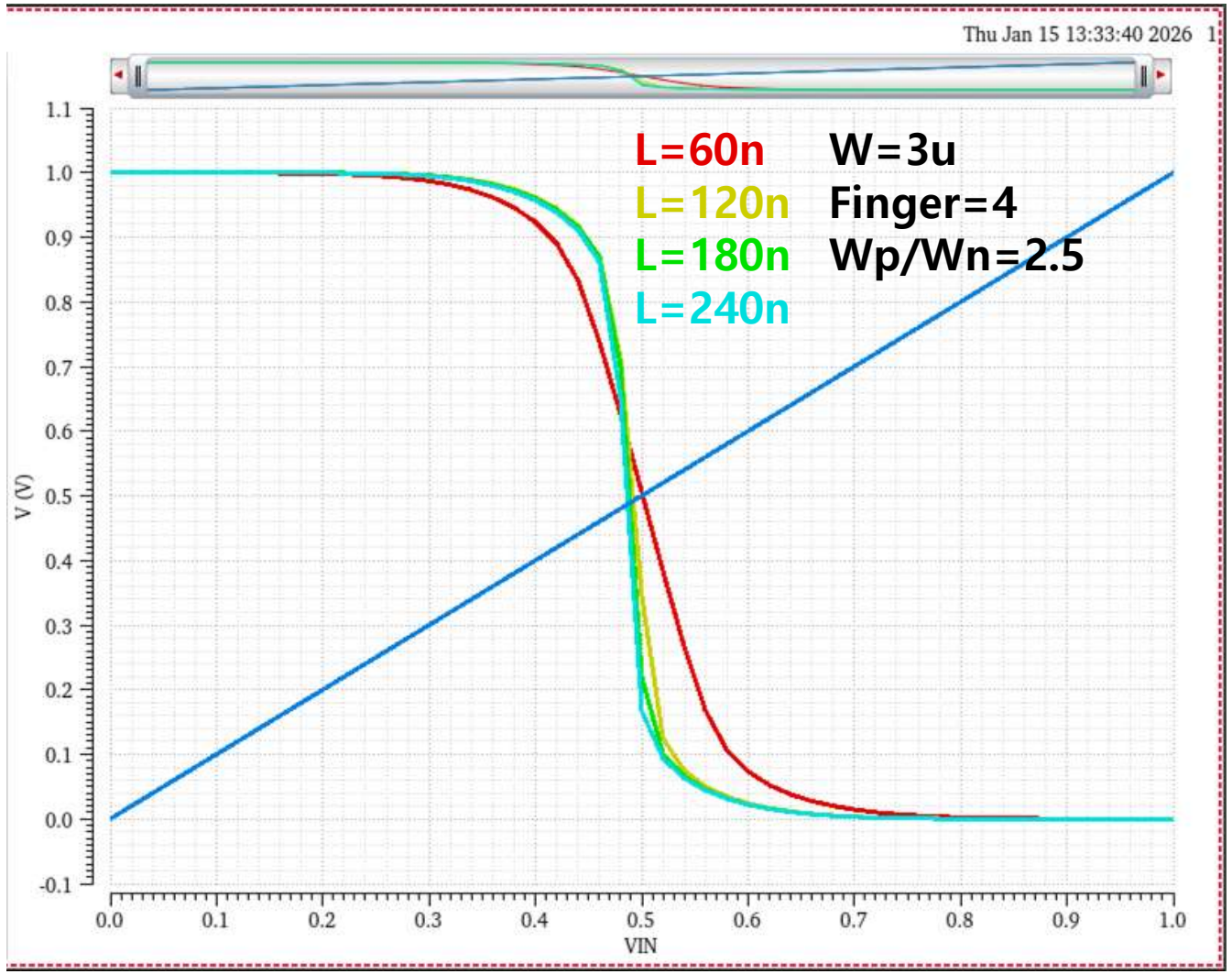


Finger In VIN Sweep

⇒ 인버터에서 mosfet의 finger 수는 영향 없음

⇒ ROSC에서 mosfet의 finger 수 결정

Length



Length In VIN Sweep

⇒ l=60n일 때, 스위칭이 제일 빠름

⇒ 전류 구동 능력 향상

⇒ 기생 cap 감소

너무 줄이면

⇒ 문턱 전압 저하

⇒ 드레인 전위장벽 저하

⇒ 누설전류

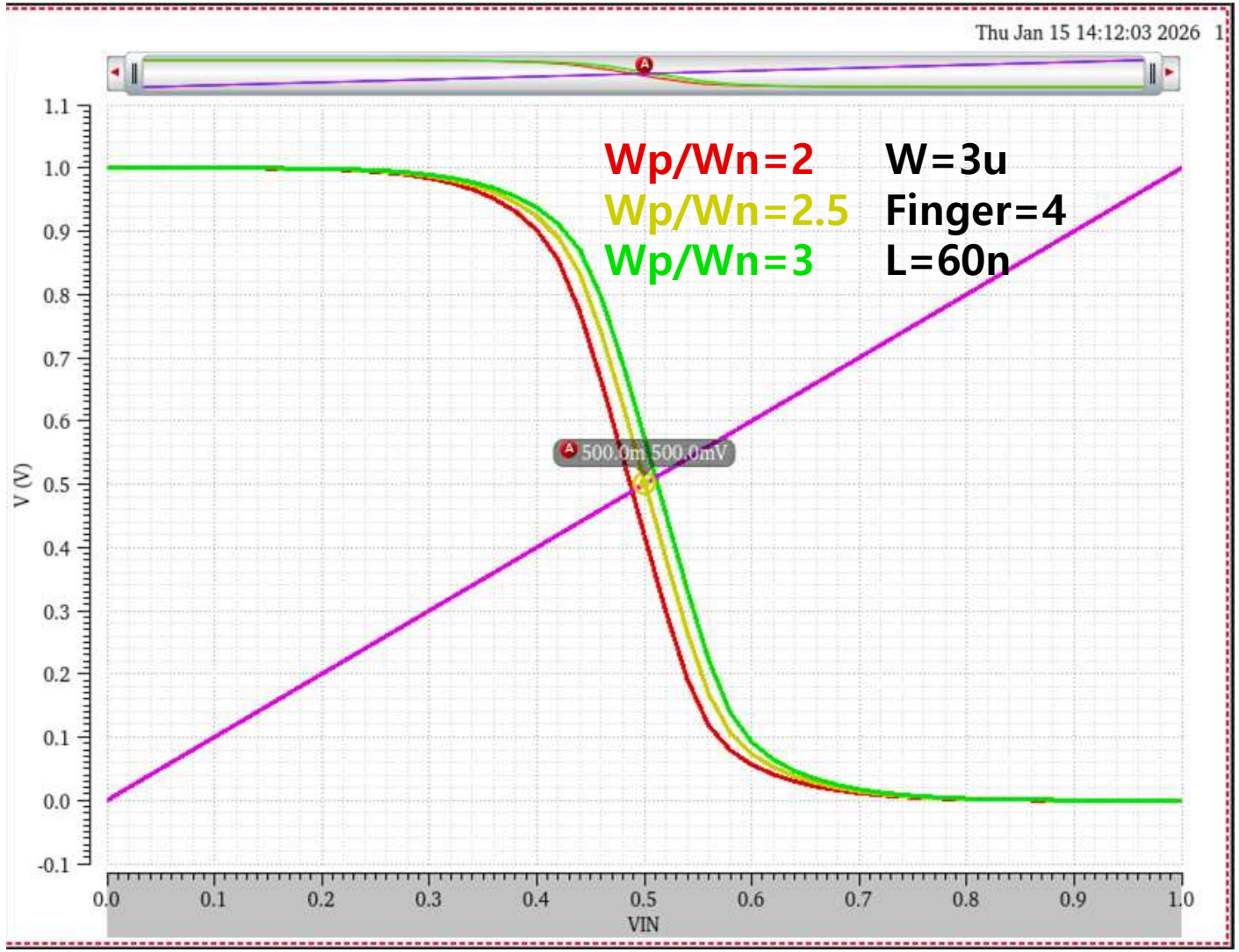
⇒ 펀치스루

⇒ S,D 공핍층 닿음

⇒ 열 전자 효과

⇒ 전계 효과 증가

Wp/Wn



Wp/Wn In VIN Sweep

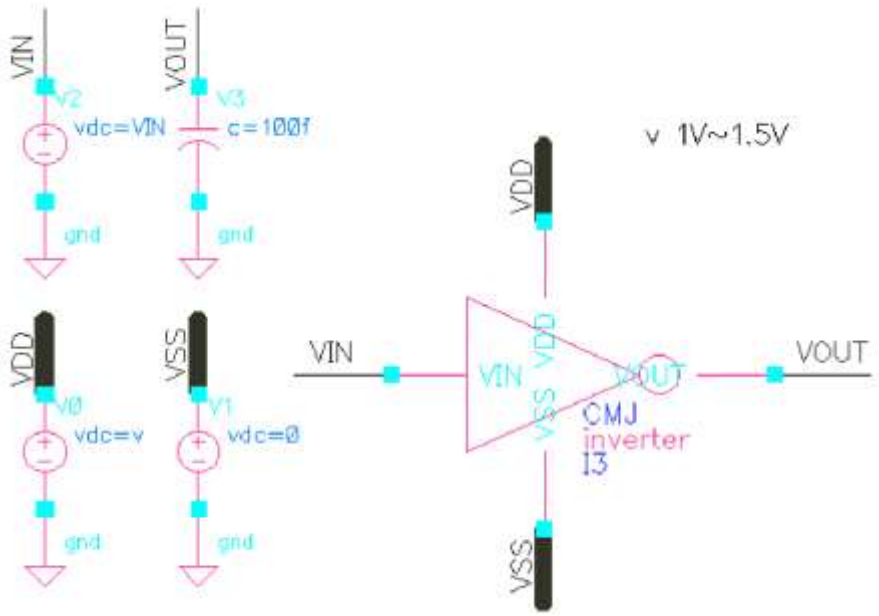
⇒ VDD 상관 없이 Wp/Wn=2.5일 때,

스위칭 전압이 상승/하강 시간이 대칭

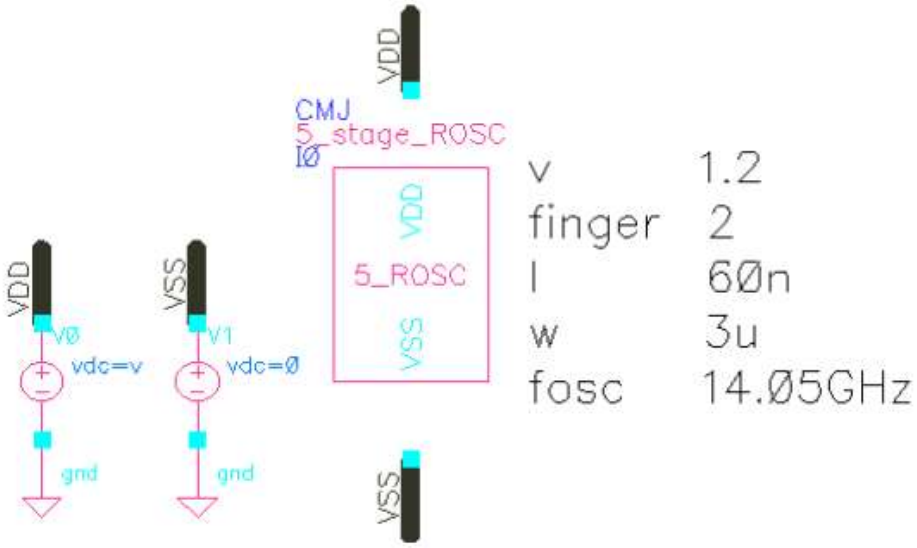
⇒ 기생 cap이 커져

실제 최적의 값이 아닐 수 있음

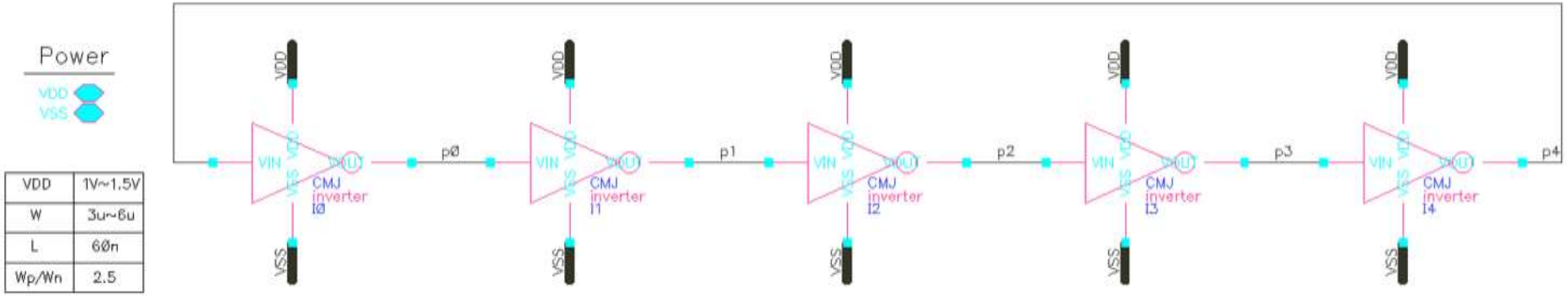
schematic



testbench



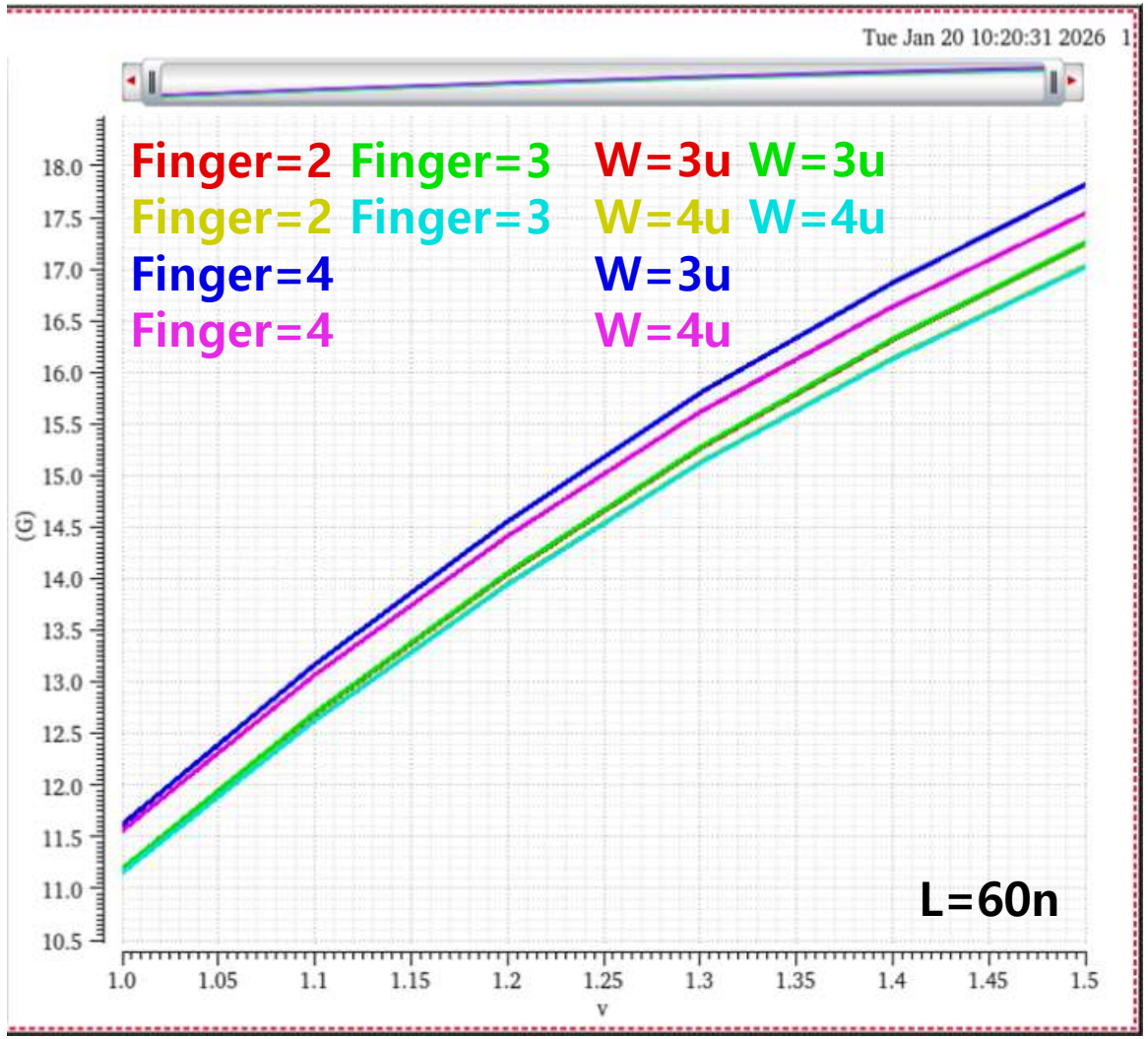
testbench



VDD	1V~1.5V
W	3u~6u
L	60n
Wp/Wn	2.5

schematic

Finger, Width



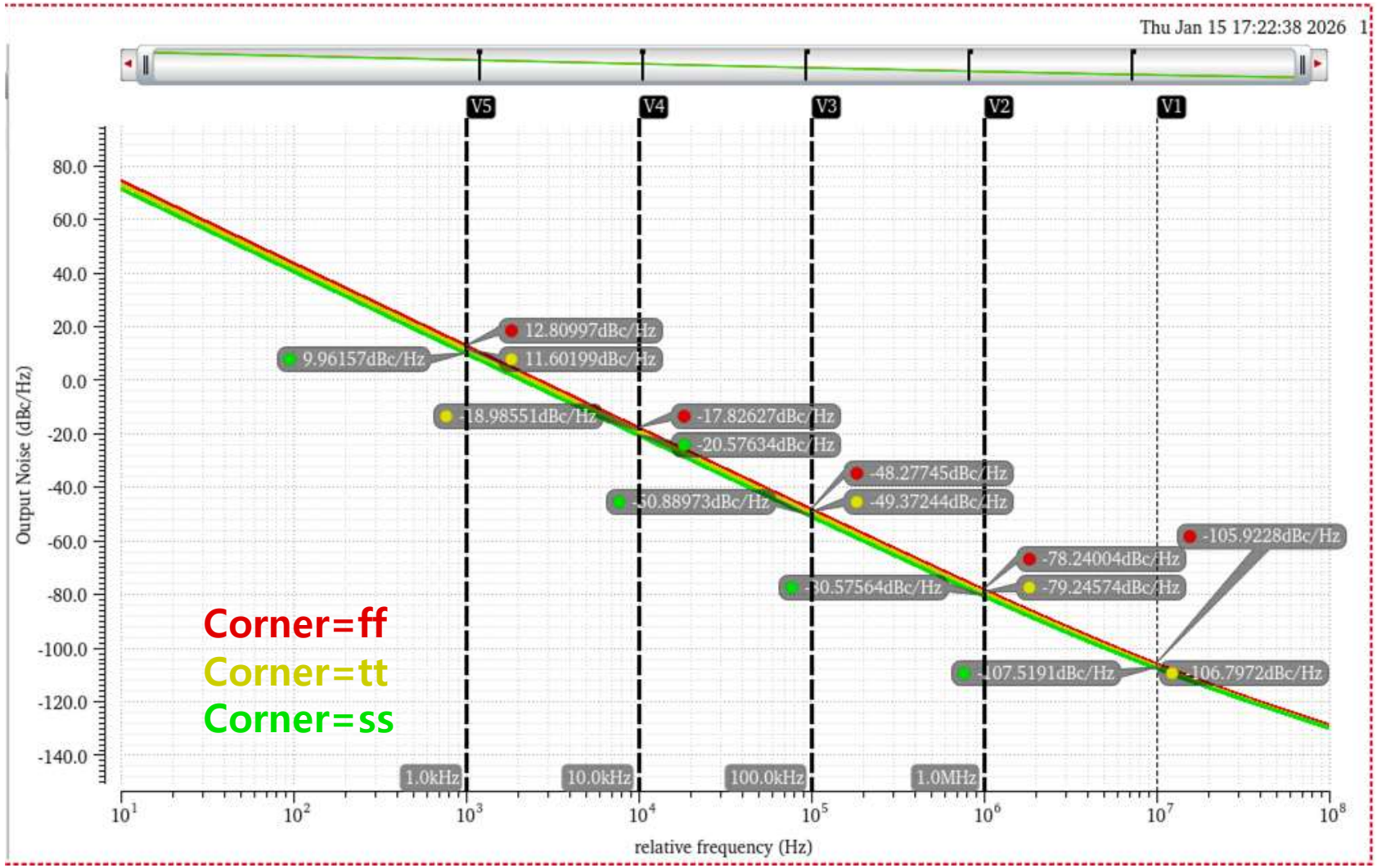
Finger, Width In VDD Sweep

⇒ Finger 4가 제일 발진주파수가 큼

⇒ 면적을 생각하면 finger 2가 최적

⇒ W 3u가 최적

Phase Noise



Phase Noise

FoM

FoM	|PN|+20log(fosc/df)-10log(Pdc/1mW)
|PN|	rfOutputNoise("dBc/Hz" ?result "pnoise_pm" ?noiseConvention "SSB")
fosc	harmonic(xval(getData("I0.net1" ?result "pss_fd")) '1)
df	Offset Frequency
Idc	rms(/IPRB0/PLUS)
Pdc/1mW	Idc*VDD/1mW

⇒ Figure of Merit

⇒ 높을수록 좋음

⇒ Periodic Static State

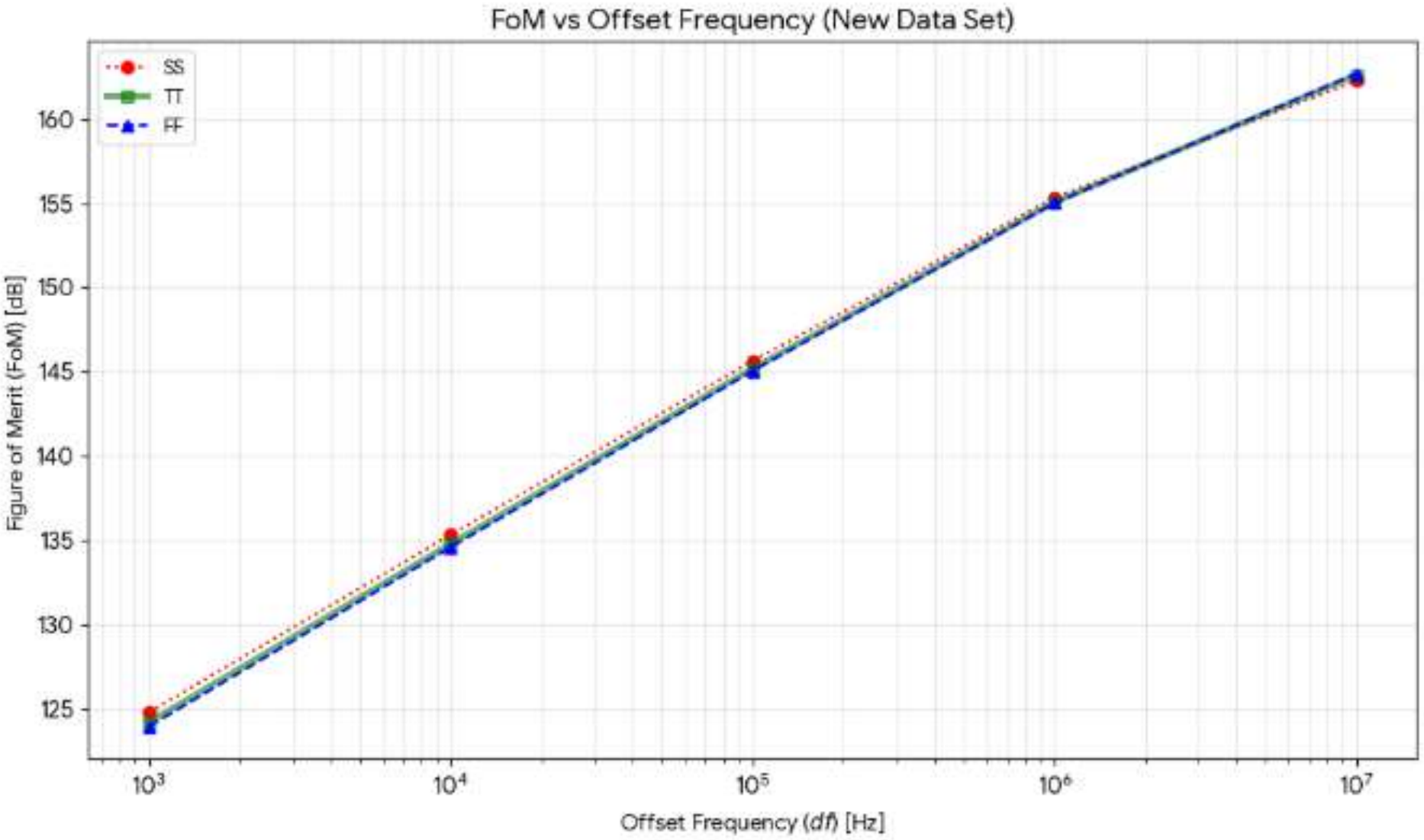
⇒ 회로가 일정하게

발진하는 안정된 한 주기

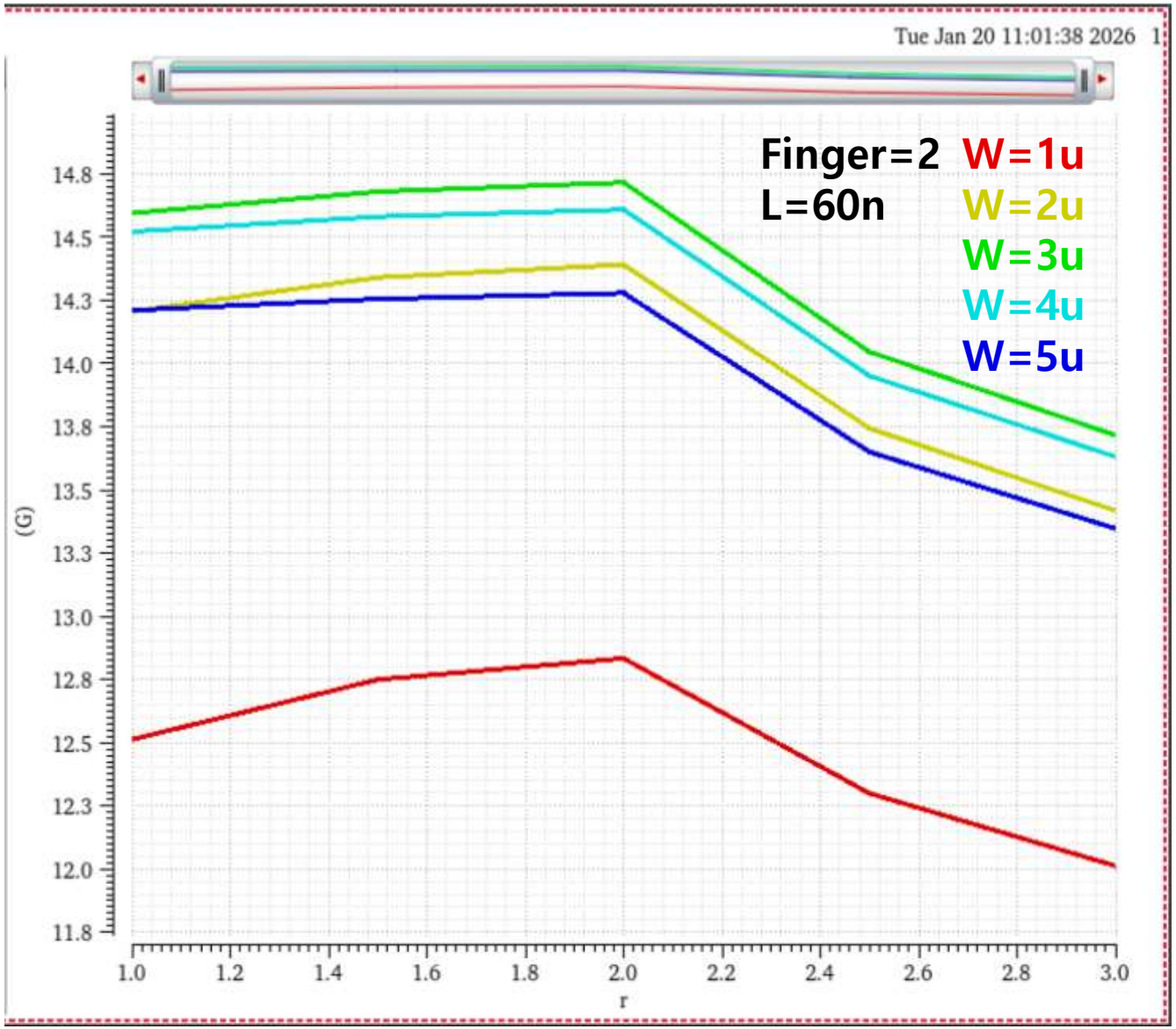
FoM

FoM = |PN|+20log(fosc/df)-10log(Pdc/1mW)		|PN|	fosc	df	Idc	Pdc/1mW
ss	144.7057068	9.96157	1.12E+10	1.00E+03	3.475	4.17
tt	147.405005	11.60199	1.41E+10	1.00E+03	4.33	5.196
ff	149.5732928	12.80997	1.74E+10	1.00E+03	5.316	6.3792
ss	135.3204768	20.57634	1.12E+10	1.00E+04	3.475	4.17
tt	134.788525	18.98551	1.41E+10	1.00E+04	4.33	5.196
ff	134.5895928	17.82627	1.74E+10	1.00E+04	5.316	6.3792
ss	145.6338668	50.88973	1.12E+10	1.00E+05	3.475	4.17
tt	145.175455	49.37244	1.41E+10	1.00E+05	4.33	5.196
ff	145.0407728	48.27745	1.74E+10	1.00E+05	5.316	6.3792
ss	155.3197768	80.57564	1.12E+10	1.00E+06	3.475	4.17
tt	155.048755	79.24574	1.41E+10	1.00E+06	4.33	5.196
ff	155.0033628	78.24004	1.74E+10	1.00E+06	5.316	6.3792
ss	162.2632368	107.5191	1.12E+10	1.00E+07	3.475	4.17
tt	162.600215	106.7972	1.41E+10	1.00E+07	4.33	5.196
ff	162.6861228	105.9228	1.74E+10	1.00E+07	5.316	6.3792

FoM


FoM vs Offset Frequency (New Data Set)
SS
TT
FF
160
155
150
145
140
135
130
125
Figure of Merit (FoM) [dB]
10^3
10^4
10^5
10^6
10^7
Offset Frequency (df) [Hz]

Wp/Wn, Width



Wp/Wn, Width In VDD Sweep

⇒ 슬라이드 6에 따른 기생 cap 때문에

Wp/Wn 비율이 맞지 않아도

발진주파수가 높아짐

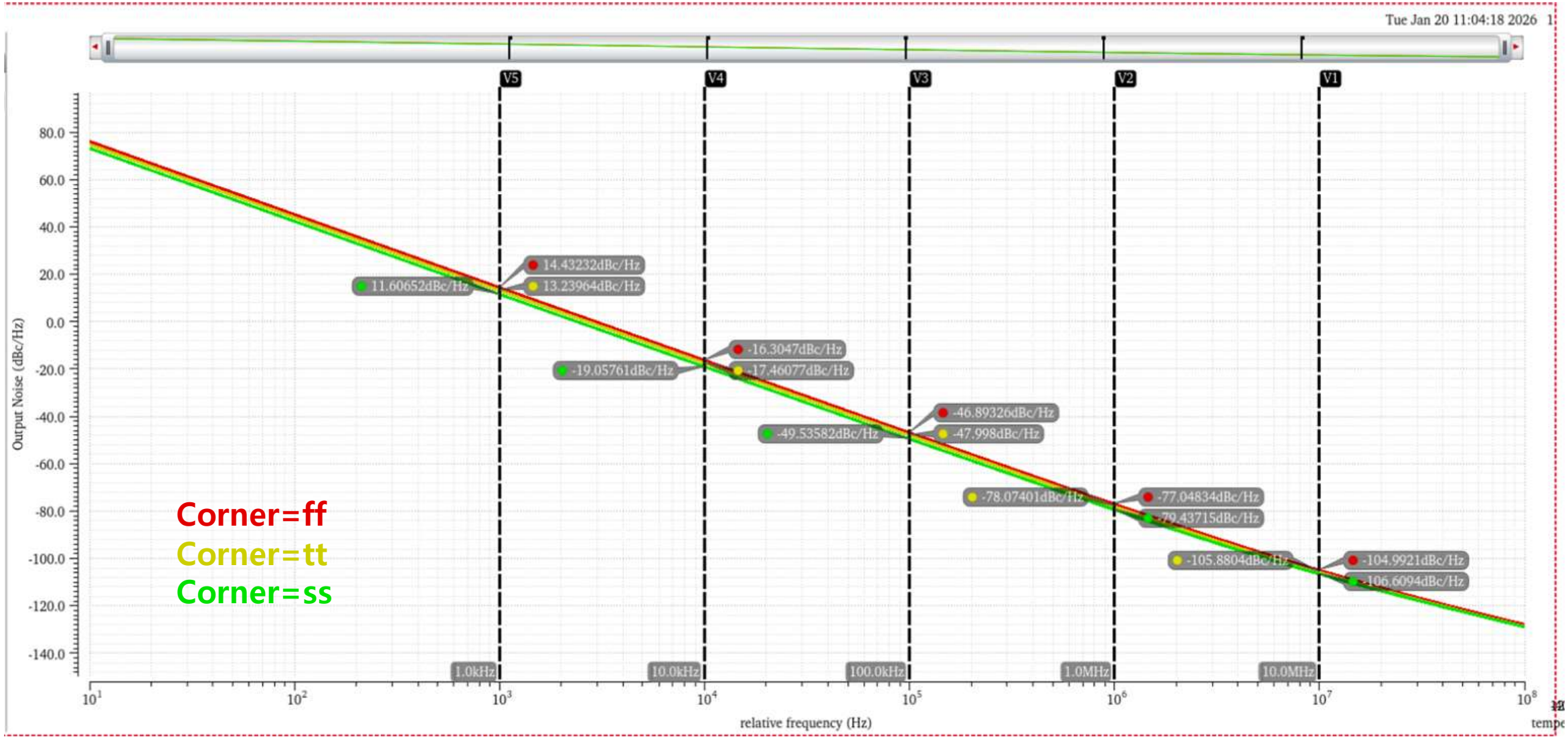
⇒ finger에서 2배와 width에서의 2배는

기생 cap과 width로 부터 생기는 cap 등

여러가지의 이유로

발진주파수의 값이 달라짐

Phase Noise



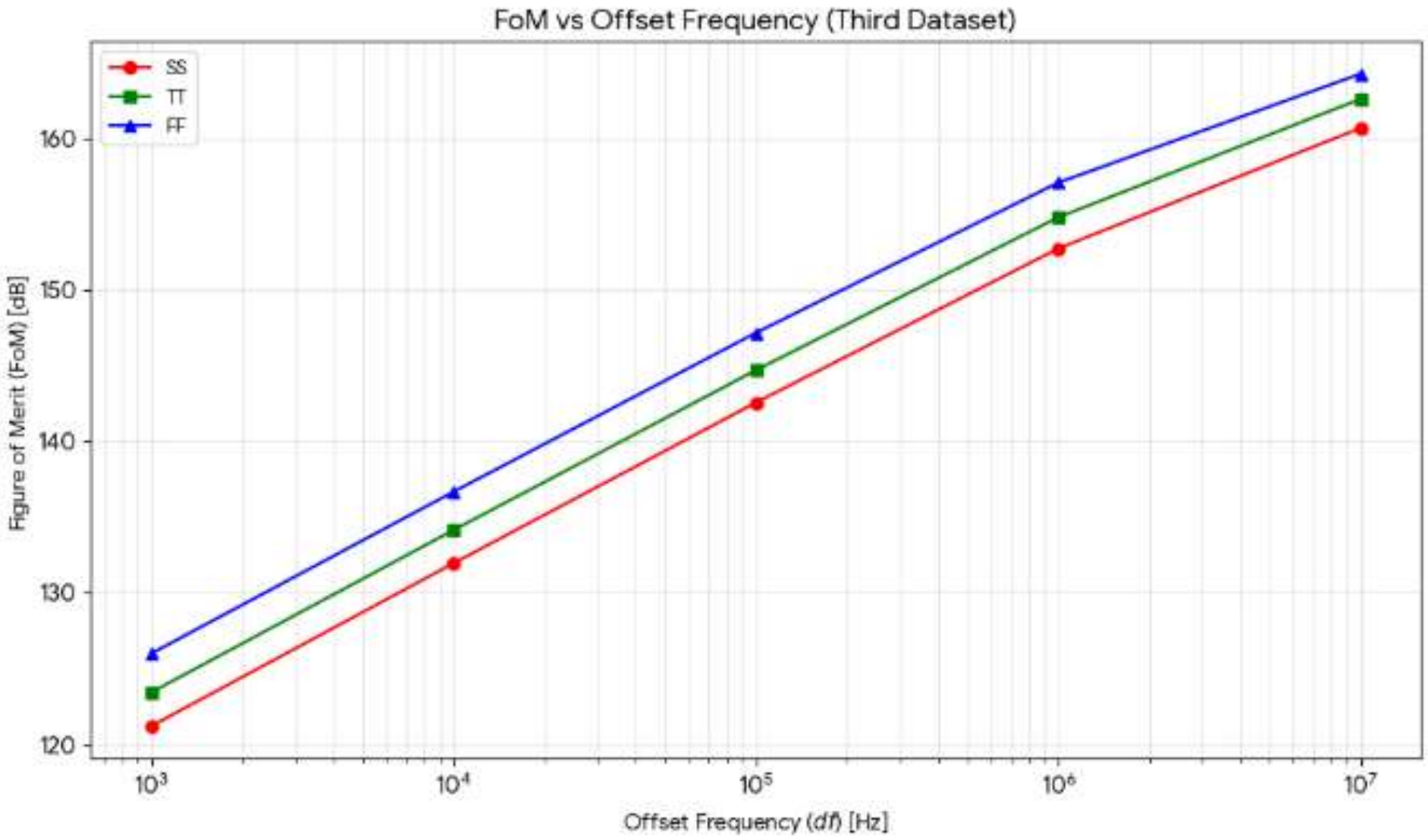
Finger=1.5일 때

FoM

	FoM	|PN|	fosc	df	Idc	Pdc/1mW
ss	150.0932817	14.43232	1.17E+10	1.00E+03	3.114	3.7368
tt	149.9302863	13.23964	1.47E+10	1.00E+03	3.874	4.6488
ff	149.225091	11.60652	1.81E+10	1.00E+03	4.745	5.694
ss	131.9656617	16.3047	1.17E+10	1.00E+04	3.114	3.7368
tt	134.1514163	17.46077	1.47E+10	1.00E+04	3.874	4.6488
ff	136.676181	19.05761	1.81E+10	1.00E+04	4.745	5.694
ss	142.5542217	46.89326	1.17E+10	1.00E+05	3.114	3.7368
tt	144.6886463	47.998	1.47E+10	1.00E+05	3.874	4.6488
ff	147.154391	49.53582	1.81E+10	1.00E+05	4.745	5.694
ss	152.7093017	77.04834	1.17E+10	1.00E+06	3.114	3.7368
tt	154.7646563	78.07401	1.47E+10	1.00E+06	3.874	4.6488
ff	157.055721	79.43715	1.81E+10	1.00E+06	4.745	5.694
ss	160.6530617	104.9921	1.17E+10	1.00E+07	3.114	3.7368
tt	162.5710463	105.8804	1.47E+10	1.00E+07	3.874	4.6488
ff	164.227971	106.6094	1.81E+10	1.00E+07	4.745	5.694

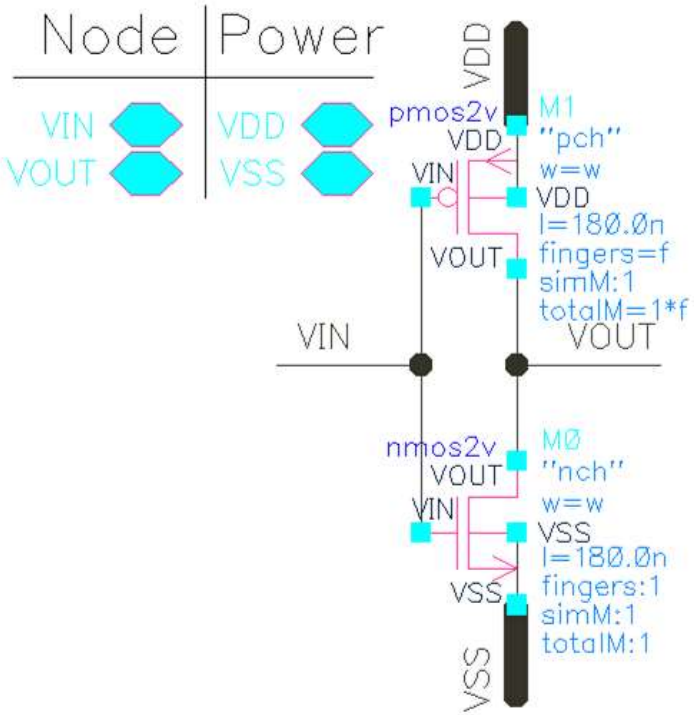
Finger=1.5일 때

FoM

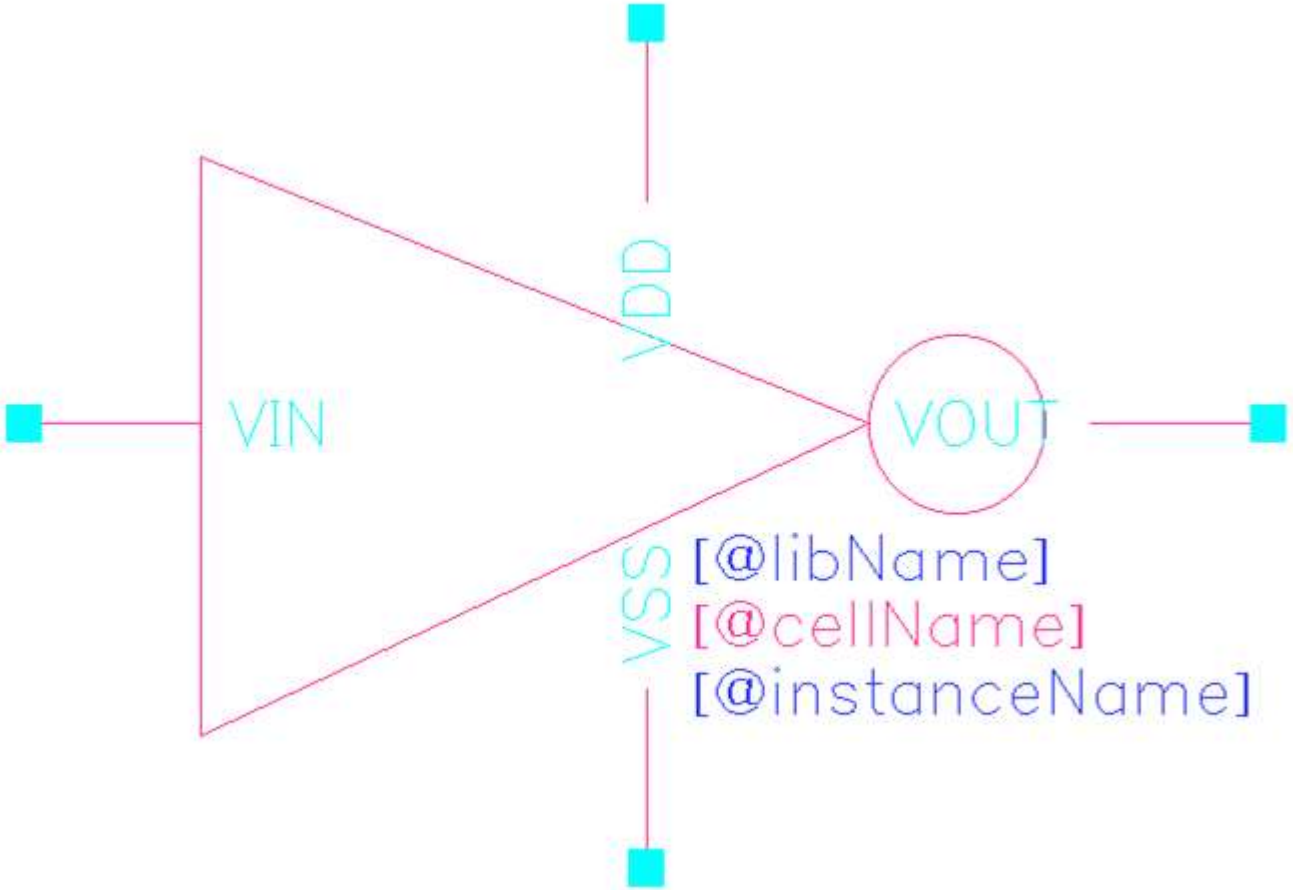


Finger=1.5일 때

180nm Schematic

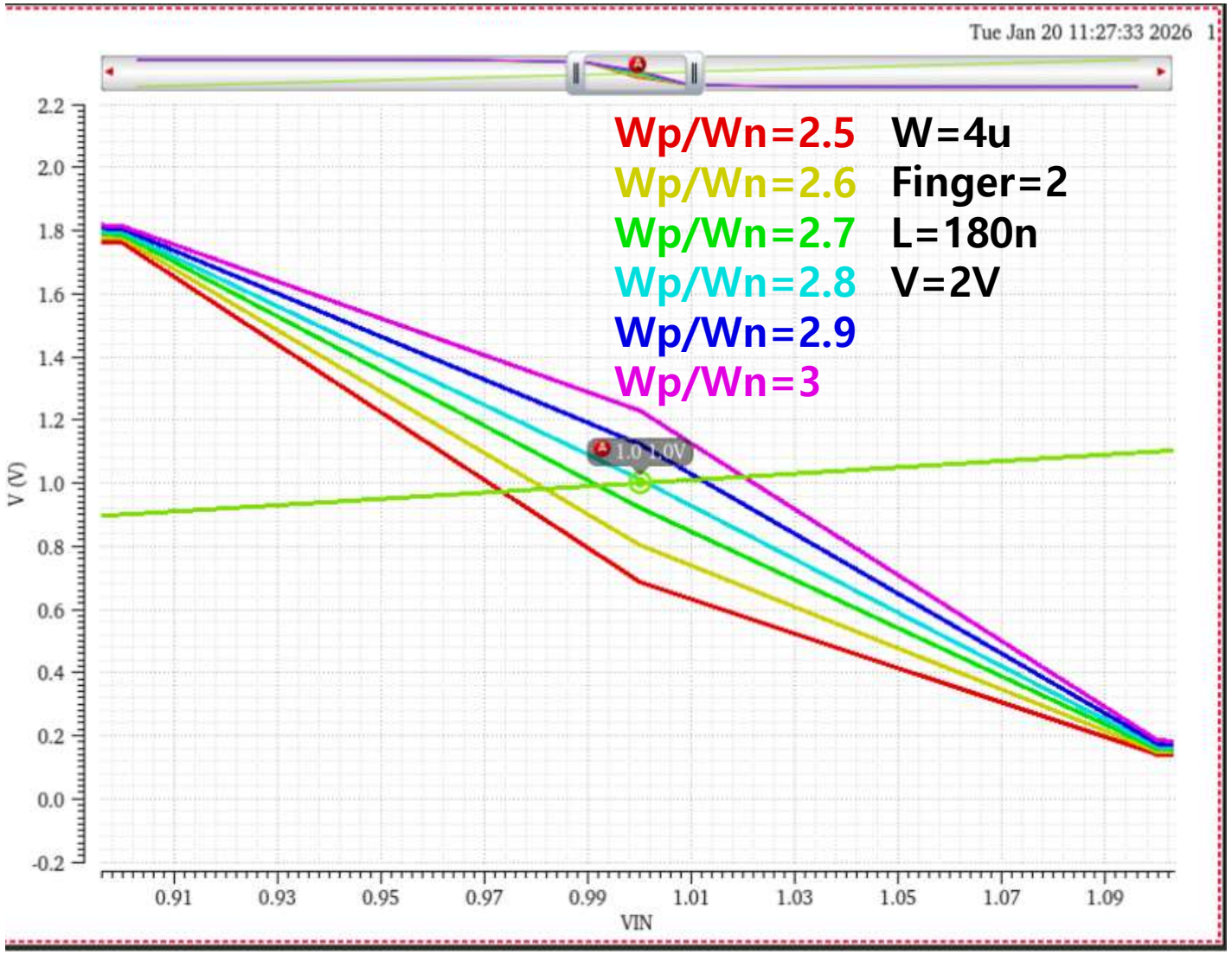


schematic



symbol

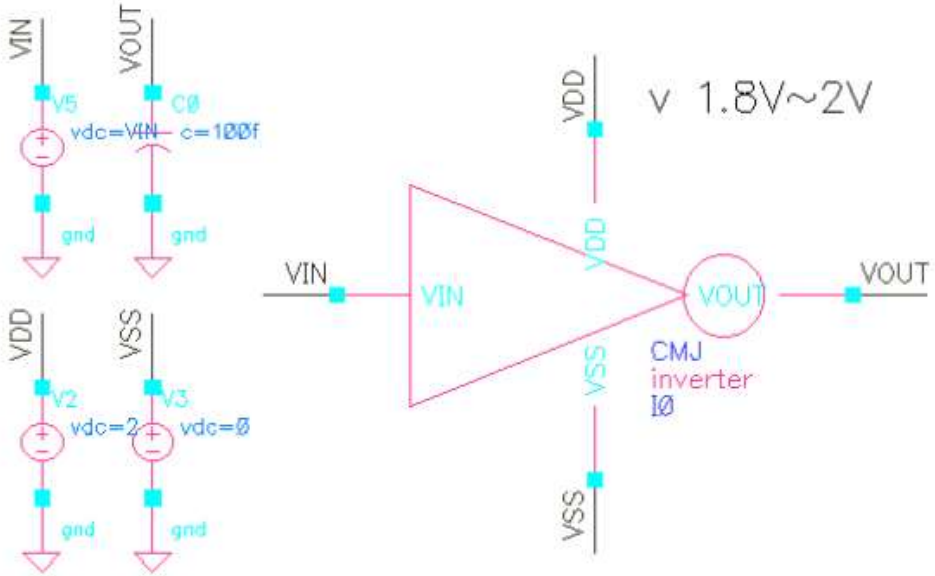
Wp/Wn



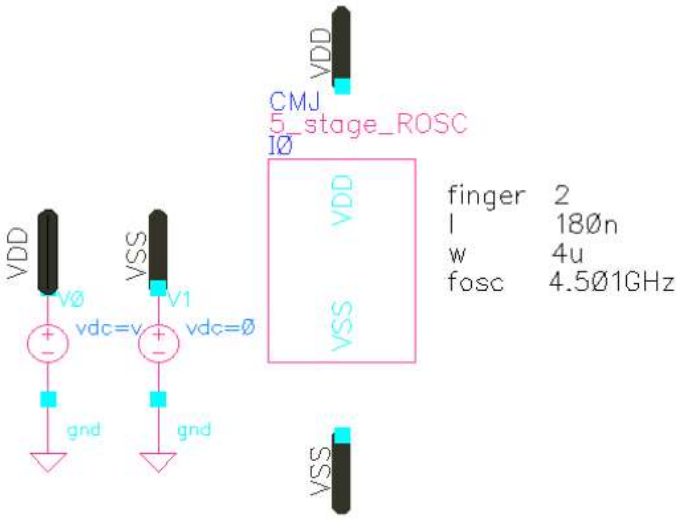
Wp/Wn In VIN Sweep

=> 대략 Wp/Wn=2.7~2.9이 최적

schematic



testbench

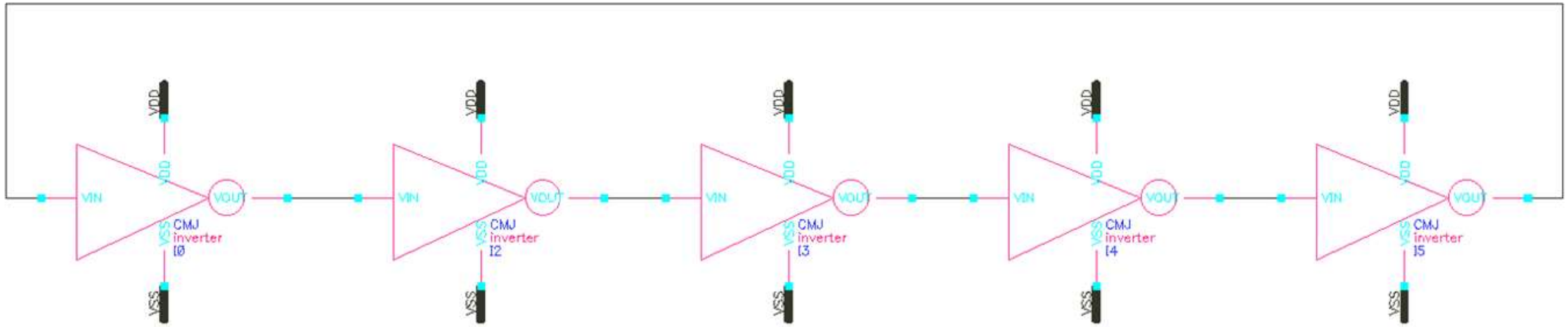


testbench

Power

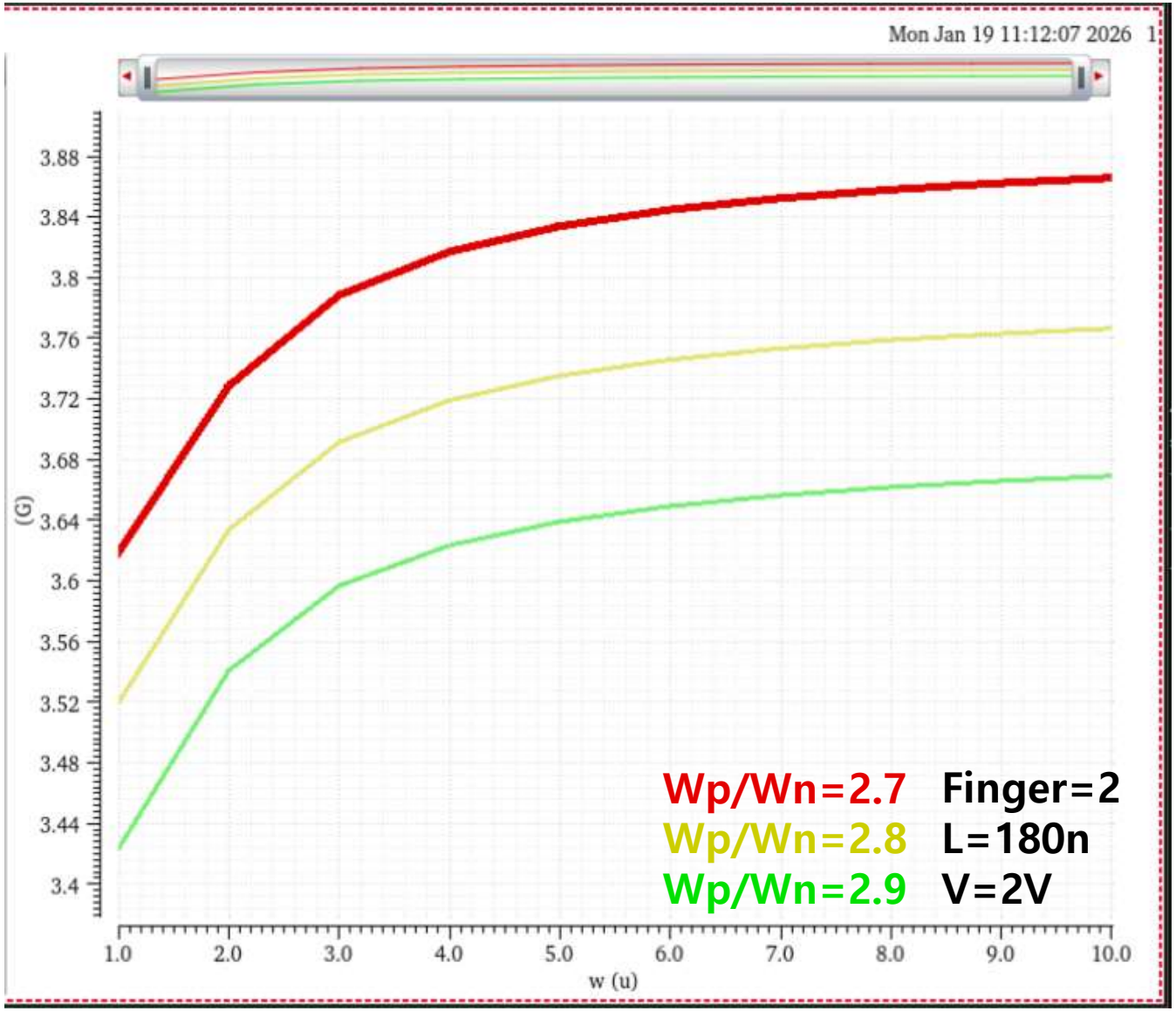
VDD
VSS

VDD	1.8V~2.5V
W	2u~5u
L	180n
Wp/Wn	2.7~2.8



schematic

Wp/Wn

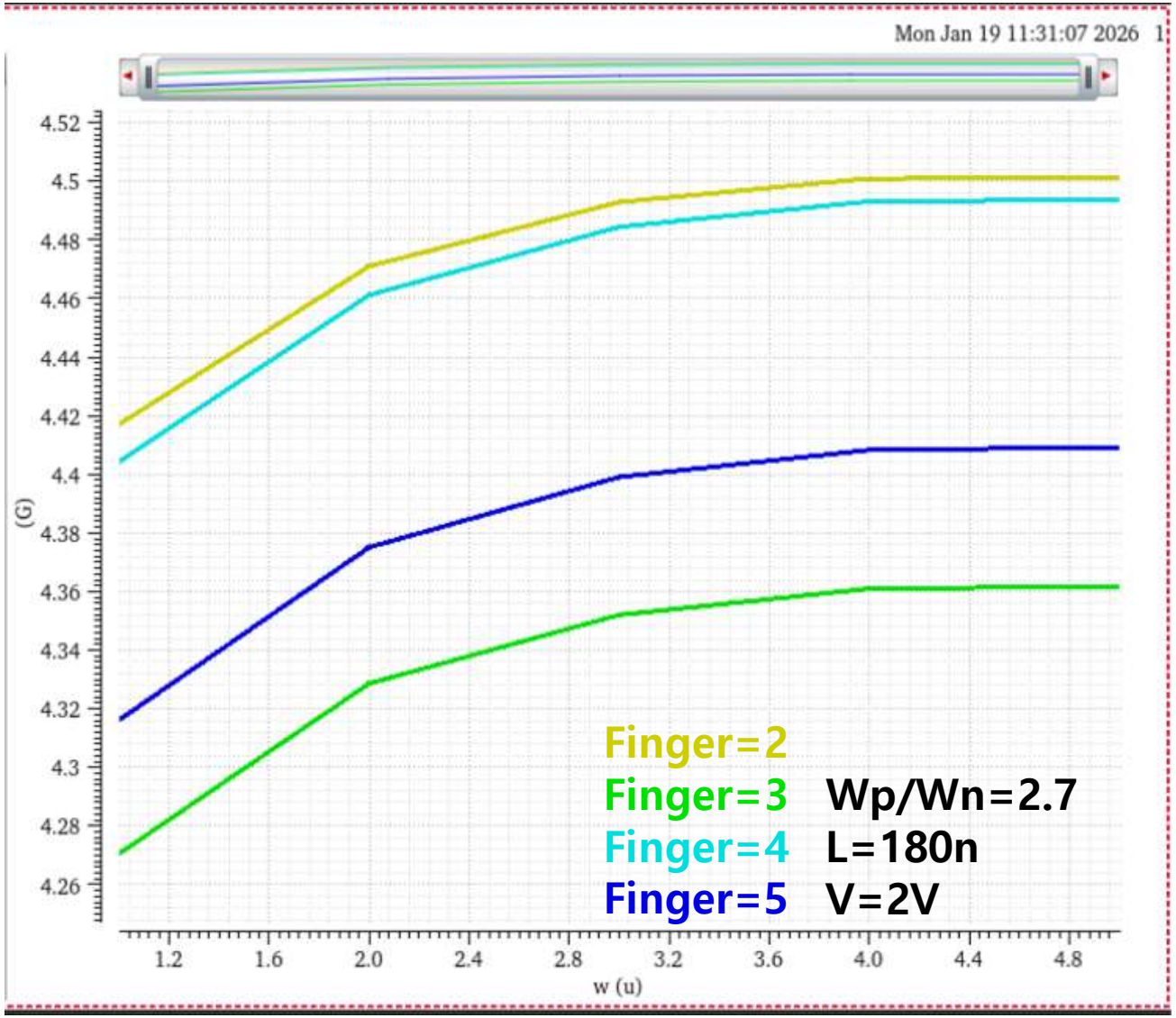


Wp/Wn in W Sweep

⇒ 소자의 크기가 작을수록

발진주파수 상승

Finger



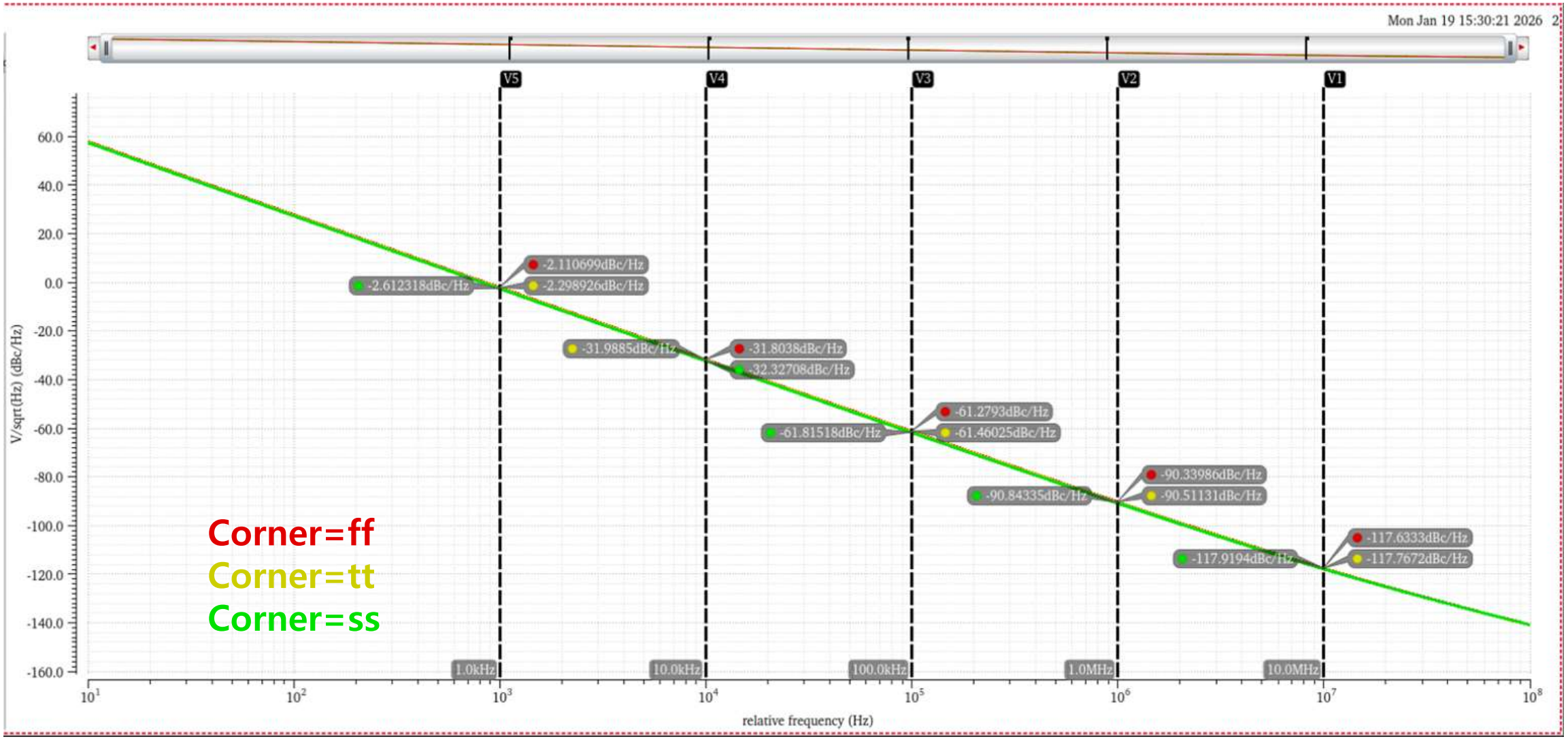
Finger in W Sweep

⇒ Finger는 짝수일 때가 좋음

⇒ D, S 영역을 공유해서

불필요한 기생 cap을 줄임

Phase Noise


Mon Jan 19 15:30:21 2026
V5
V4
V3
V2
V1
-2.110699dBc/Hz
-2.612318dBc/Hz
-2.298926dBc/Hz
-31.9885dBc/Hz
-31.8038dBc/Hz
-32.32708dBc/Hz
-61.2793dBc/Hz
-61.81518dBc/Hz
-61.46025dBc/Hz
-90.33986dBc/Hz
-90.84335dBc/Hz
-90.51131dBc/Hz
-117.6333dBc/Hz
-117.9194dBc/Hz
-117.7672dBc/Hz
1.0kHz
10.0kHz
100.0kHz
1.0MHz
10.0MHz
Corner=ff
Corner=tt
Corner=ss
V/sqrt(Hz) (dBc/Hz)
relative frequency (Hz)

FoM

FoM = |PN|+20log(fosc/df)-10log(Pdc/1mW)		|PN|	fosc	df	Idc	Pdc/1mW
ss	125.6501055	2.612318	3.88E+09	1.00E+03	3.736	7.472
tt	126.0985955	2.298926	4.50E+09	1.00E+03	4.223	8.446
ff	126.4460362	2.110699	5.01E+09	1.00E+03	4.625	9.25
ss	135.3648675	32.32708	3.88E+09	1.00E+04	3.736	7.472
tt	135.7881695	31.9885	4.50E+09	1.00E+04	4.223	8.446
ff	136.1391372	31.8038	5.01E+09	1.00E+04	4.625	9.25
ss	144.8529675	61.81518	3.88E+09	1.00E+05	3.736	7.472
tt	145.2599195	61.46025	4.50E+09	1.00E+05	4.223	8.446
ff	145.6146372	61.2793	5.01E+09	1.00E+05	4.625	9.25
ss	153.8811375	90.84335	3.88E+09	1.00E+06	3.736	7.472
tt	154.3109795	90.51131	4.50E+09	1.00E+06	4.223	8.446
ff	154.6751972	90.33986	5.01E+09	1.00E+06	4.625	9.25
ss	160.9571875	117.9194	3.88E+09	1.00E+07	3.736	7.472
tt	161.5668695	117.7672	4.50E+09	1.00E+07	4.223	8.446
ff	161.9686372	117.6333	5.01E+09	1.00E+07	4.625	9.25

FoM

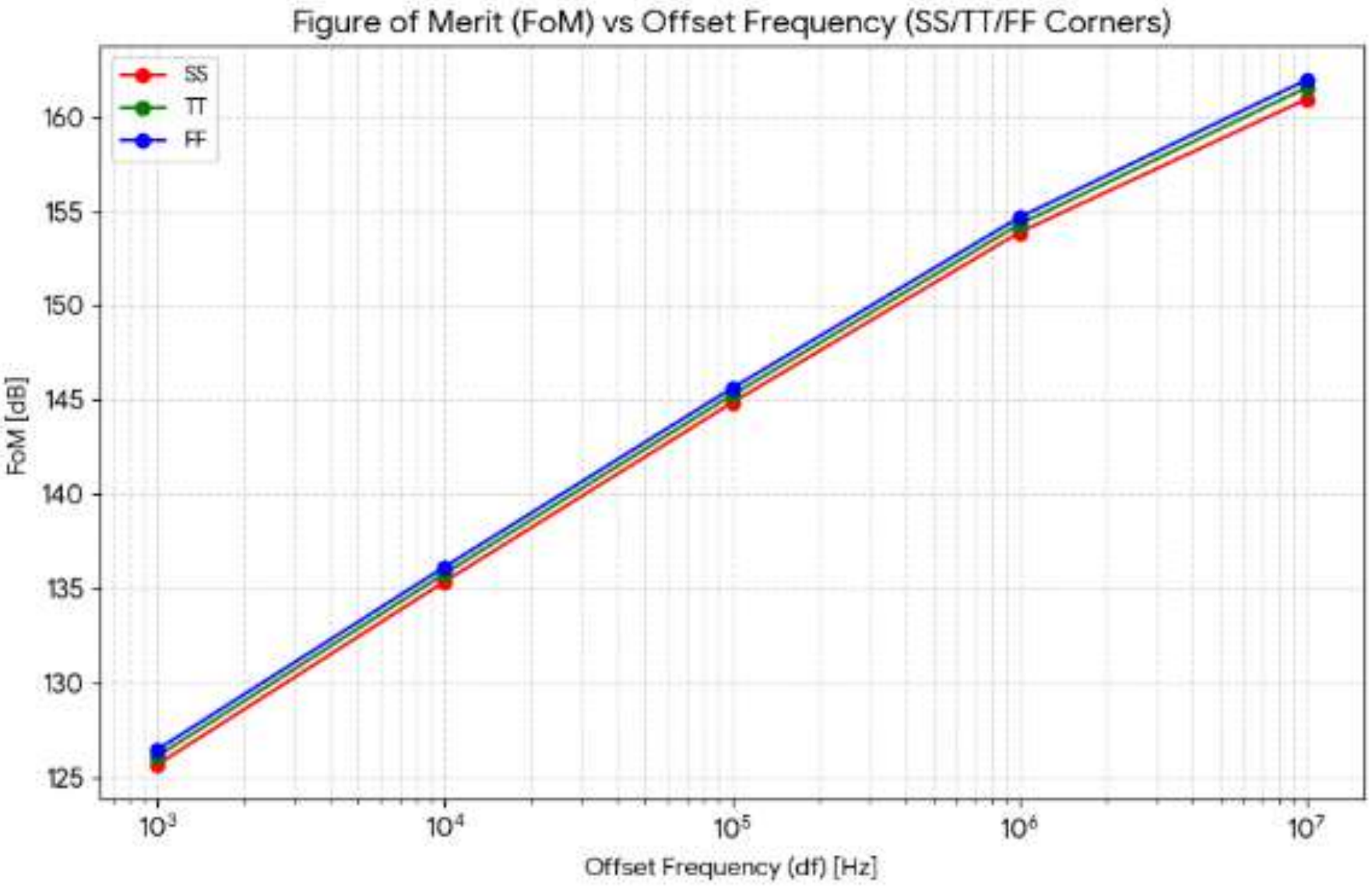


Figure of Merit (FoM) vs Offset Frequency (SS/TT/FF Corners)
SS
TT
FF
160
155
150
145
140
135
130
125
FoM [dB]
10^3
10^4
10^5
10^6
10^7
Offset Frequency (df) [Hz]

Summary

65nm	180nm
125.6501055	124.7825668
126.0985955	124.201025
126.4460362	123.9533528
135.3648675	135.3204768
135.7881695	134.788525
136.1391372	134.5895928
144.8529675	145.6338668
145.2599195	145.175455
145.6146372	145.0407728
153.8811375	155.3197768
154.3109795	155.048755
154.6751972	155.0033628
160.9571875	162.2632368
161.5668695	162.600215
161.9686372	162.6861228

65nm와 180nm 사이의 FoM의 차이는 크게 느끼지 못하였다

⇒ 만일, FoM이 비슷한 스펙이라고 하면 발진주파수가 높고

면적이 훨씬 작은 tsmc65nm를 쓰는 게 더 유용하다는 생각이 든다

65nm	180nm
3.878G	11.15G
4.501G	14.06G
5.01G	17.4G

★위에서부터 ss, tt, ff 순